

Rising Sea (Week 2): Adjoints and Sheaves

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I will follow *The Rising Sea*'s structure. I am definitely not flawless, so please give feedback.

Lecture overview

- **Adjoints:** natural Hom-set bijections; unit and counit; tensor–Hom, extension–restriction, Frobenius reciprocity, and localisation.
- **Sheaves:** smooth functions as the motivating example; germs and stalks; presheaves and the sheaf axioms; the espace étalé viewpoint.
- **Local algebra:** ringed spaces, $\mathcal{O}_X$ -modules, and stalks as the algebra attached to a point.

1 Adjoints (RS 1.4)

Say: A universal property determines an object up to unique isomorphism. An adjunction is the corresponding functorial phenomenon: a left adjoint determines its right adjoint up to natural isomorphism, and conversely.

Definition and naturality

Definition 1.1 (Adjunction). Let $F : \mathcal{A} \longrightarrow \mathcal{B}$, and $G : \mathcal{B} \longrightarrow \mathcal{A}$. be covariant functors. We say that F is **left-adjoint** to G , or equivalently that G is **right-adjoint** to F , if for every $A \in \mathcal{A}$ and $B \in \mathcal{B}$ there is a natural bijection

$$\tau_{A,B} : \text{Mor}_{\mathcal{B}}(F(A), B) \xrightarrow{\sim} \text{Mor}_{\mathcal{A}}(A, G(B)).$$

We write $F \dashv G$.

Key point. The bijection is not just a separate bijection for each (A, B) . It must respect precomposition in A and postcomposition in B .

Naturality.

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A'), B) & \xrightarrow{(Ff)^*} & \text{Mor}_{\mathcal{B}}(F(A), B) \\ \tau_{A',B} \downarrow & & \downarrow \tau_{A,B} \\ \text{Mor}_{\mathcal{A}}(A', G(B)) & \xrightarrow{f^*} & \text{Mor}_{\mathcal{A}}(A, G(B)) \end{array} \qquad \begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), B) & \xrightarrow{g_*} & \text{Mor}_{\mathcal{B}}(F(A), B') \\ \tau_{A,B} \downarrow & & \downarrow \tau_{A,B'} \\ \text{Mor}_{\mathcal{A}}(A, G(B)) & \xrightarrow{(Gg)_*} & \text{Mor}_{\mathcal{A}}(A, G(B')) \end{array}$$

for $f : A \rightarrow A'$ and $g : B \rightarrow B'$.

- *precomposition*: $f^*(h) = h \circ f$, $(Ff)^*(k) = k \circ Ff$
- *postcomposition*: $g_*(h) = g \circ h$, $(Gg)_*(k) = Gg \circ k$

Unit and counit

Key point. Question: Can we extract canonical morphisms from the adjunction bijection itself?

Lemma 1.2 (Unit and counit). *Suppose $F \dashv G$. For every $A \in \mathcal{A}$ and $B \in \mathcal{B}$ there are canonical morphisms*

$$\eta_A : A \longrightarrow GF(A), \quad \varepsilon_B : FG(B) \longrightarrow B.$$

*They are called the **unit** and **counit** of the adjunction.*

Proof. Set $B = F(A)$ in the adjunction bijection:

$$\tau_{A, F(A)} : \text{Mor}_{\mathcal{B}}(F(A), F(A)) \xrightarrow{\sim} \text{Mor}_{\mathcal{A}}(A, GF(A)).$$

Define $\eta_A := \tau_{A, F(A)}(\text{id}_{F(A)})$. For the counit, set $A = G(B)$ and use the inverse bijection:

$$\tau_{G(B), B}^{-1} : \text{Mor}_{\mathcal{A}}(G(B), G(B)) \xrightarrow{\sim} \text{Mor}_{\mathcal{B}}(FG(B), B).$$

and define $\varepsilon_B := \tau_{G(B), B}^{-1}(\text{id}_{G(B)})$. □

Say: The unit is the adjoint transpose of an identity on the F -side; the counit is the adjoint transpose of an identity on the G -side.

Lemma 1.3 (Recovering the adjoint transpose). *Suppose $F \dashv G$. Then:*

1. *for every $g : F(A) \rightarrow B$, $\tau_{A, B}(g) = G(g) \circ \eta_A$,*
2. *for every $f : A \rightarrow G(B)$, $\tau_{A, B}^{-1}(f) = \varepsilon_B \circ F(f)$.*

Proof. For (1), use naturality in the second variable for the morphism $g : F(A) \rightarrow B$:

$$\begin{array}{ccc} \text{Mor}_{\mathcal{B}}(F(A), F(A)) & \xrightarrow{g_*} & \text{Mor}_{\mathcal{B}}(F(A), B) \\ \tau_{A, F(A)} \downarrow & & \downarrow \tau_{A, B} \\ \text{Mor}_{\mathcal{A}}(A, GF(A)) & \xrightarrow{(Gg)_*} & \text{Mor}_{\mathcal{A}}(A, G(B)). \end{array}$$

Evaluate at $\text{id}_{F(A)}$. Along the top, $g_*(\text{id}_{F(A)}) = g$. Along the left, $\tau_{A, F(A)}(\text{id}_{F(A)}) = \eta_A$. Hence $\tau_{A, B}(g) = G(g) \circ \eta_A$.

The second statement is the same argument using naturality in the first variable, starting from $\tau_{G(B), B}(\varepsilon_B) = \text{id}_{G(B)}$. □

Examples of adjunctions

Say: Quick example. The next example is important and will reappear in localisation.

Example 1.4 (Tensor–Hom adjunction). Let A be a ring and N an A -module. Then

$$- \otimes_A N : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_A$$

is left-adjoint to

$$\mathrm{Hom}_A(N, -) : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_A.$$

Equivalently,

$$\mathrm{Hom}_A(M \otimes_A N, P) \cong \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P))$$

naturally in M and P .

Example 1.5 (Extension and restriction of scalars). Let $\varphi : B \rightarrow A$ be a morphism of rings. Every A -module M becomes a B -module by $b \cdot m := \varphi(b)m$. This defines the restriction/extension of scalars functors

$$\mathrm{Res}_B^A : \mathbf{Mod}_A \longrightarrow \mathbf{Mod}_B, \quad - \otimes_B A : \mathbf{Mod}_B \longrightarrow \mathbf{Mod}_A.$$

Then for $N \in \mathbf{Mod}_B$ and $M \in \mathbf{Mod}_A$,

$$\mathrm{Hom}_A(N \otimes_B A, M) \cong \mathrm{Hom}_B(N, \mathrm{Res}_B^A M).$$

Proof. Given an A -linear map $\Phi : N \otimes_B A \rightarrow M$, define

$$\Theta(\Phi) : N \rightarrow \mathrm{Res}_B^A M, \quad \Theta(\Phi)(n) = \Phi(n \otimes 1),$$

which is B -linear.

Conversely, given a B -linear map $f : N \rightarrow \mathrm{Res}_B^A M$, define

$$\Psi(f) : N \otimes_B A \rightarrow M, \quad \Psi(f)(n \otimes a) = a f(n),$$

which is well-defined because:

$$\Psi(f)(bn \otimes a) = a f(bn) = a \varphi(b)f(n) = \Psi(f)(n \otimes \varphi(b)a).$$

It is clearly A -linear.

Check: The maps are inverses of each other, and compatible with pre/post composition, so natural. $\square$

Example 1.6 (Frobenius reciprocity). Let $H \leq G$ be groups and k a field. Restriction gives $\mathrm{Res}_H^G : \mathbf{Rep}_k(G) \longrightarrow \mathbf{Rep}_k(H)$, while induction gives $\mathrm{Ind}_H^G : \mathbf{Rep}_k(H) \longrightarrow \mathbf{Rep}_k(G)$, where $\mathrm{Ind}_H^G(W) = kG \otimes_{kH} W$. Then

$$\mathrm{Hom}_{kG}(\mathrm{Ind}_H^G W, V) \cong \mathrm{Hom}_{kH}(W, \mathrm{Res}_H^G V).$$

Say: This is really just extension–restriction of scalars for the ring inclusion $kH \hookrightarrow kG$.

Localisation as a left adjoint

Say: This is the example I want to finish the adjoints part with, because it is both important algebraically and a preview of what a stalk is doing locally.

Let A be a ring and $S \subseteq A$ a multiplicative subset. Recall

$$S^{-1}A = \left\{ \frac{a}{s} : a \in A, s \in S \right\},$$

with canonical map $\iota : A \rightarrow S^{-1}A$, where $a \mapsto \frac{a}{1}$, and universal property

$$\begin{array}{ccc} A & \xrightarrow{\iota} & S^{-1}A \\ & \searrow f & \downarrow \exists! \tilde{f} \\ & & B \end{array}$$

Definition 1.7 (Restriction of scalars). Every $S^{-1}A$ -module N becomes an A -module via

$$a \cdot n := \frac{a}{1}n.$$

This defines the **restriction of scalars functor**

$$U : \mathbf{Mod}_{S^{-1}A} \rightarrow \mathbf{Mod}_A.$$

Remark 1.8. The functor U has two useful properties:

1. Its essential image consists precisely of those A -modules N for which multiplication by every $s \in S$,

$$s \cdot - : N \rightarrow N,$$

is an isomorphism. Indeed, on an $S^{-1}A$ -module its inverse is multiplication by $1/s$; conversely, if every s acts invertibly, the A -action extends uniquely to an $S^{-1}A$ -action.

2. U is **fully faithful**. If $\varphi : N \rightarrow N'$ is A -linear, then $\varphi(sn) = s\varphi(n)$. Since s acts invertibly, also $\varphi(s^{-1}n) = s^{-1}\varphi(n)$, and hence φ is automatically $S^{-1}A$ -linear.

Definition 1.9 (Localisation functor). Define

$$L = S^{-1}(-) : \mathbf{Mod}_A \rightarrow \mathbf{Mod}_{S^{-1}A}, \quad M \mapsto S^{-1}M.$$

For $f : M \rightarrow M'$, $S^{-1}f\left(\frac{m}{s}\right) = \frac{f(m)}{s}$. The canonical localisation map is

$$\eta_M : M \rightarrow U(S^{-1}M), \quad m \mapsto \frac{m}{1}.$$

Proposition 1.10. $S^{-1}(-) \dashv U$. i.e. $\mathrm{Hom}_{S^{-1}A}(S^{-1}M, N) \cong \mathrm{Hom}_A(M, U(N))$ naturally in M and N .

Proof. Given $\Phi : S^{-1}M \rightarrow N$, define

$$\Theta(\Phi) = \Phi \circ \eta_M, \quad \Theta(\Phi)(m) = \Phi\left(\frac{m}{1}\right).$$

Conversely, let $f : M \rightarrow U(N)$ be A -linear. Since every $s \in S$ acts invertibly on N , define

$$\tilde{f} : S^{-1}M \rightarrow N, \quad \tilde{f}\left(\frac{m}{s}\right) = s^{-1}f(m),$$

which is well-defined and $S^{-1}A$ -linear. It is the unique map satisfying $\tilde{f} \circ \eta_M = f$.

Check: inverses and naturality. □

Remark 1.11. $\mathbf{Mod}_{S^{-1}A}$ is a **reflective subcategory** of $\mathbf{Mod}_A$: localisation is the universal way of forcing every $s \in S$ to act invertibly.

Summary of examples

left adjoint	right adjoint
$- \otimes_A N$	$\mathrm{Hom}_A(N, -)$
$- \otimes_B A$	Res_B^A
Ind_H^G	Res_H^G
groupification	forgetful
$S^{-1}(-)$	U

Key point. Left adjoints tend to *freely add* structure or impose a universal construction; right adjoints tend to *forget* or retain compatible structure.

2 Sheaves (RS 2.1–2.2)

Say: Very brief history: Sheaves were introduced by Leray in the 1940s (held captive, pretended to be topologist), brought into algebraic geometry by Serre.

Motivating example: the sheaf of smooth functions

Say: A recurring principle in geometry is that a space can be studied through the functions defined on it. Global functions alone are often insufficient, so we keep track of functions on every open subset.

Let X be a smooth manifold (keep $X = \mathbb{R}^n$ in mind). For every open subset $U \subseteq X$, define $\mathcal{O}(U) := C^\infty(U, \mathbb{R})$. This is a commutative ring.

If $U \subseteq V$ are open, define the **restriction map**

$$\text{res}_{V,U} : \mathcal{O}(V) \longrightarrow \mathcal{O}(U), \quad f \longmapsto f|_U.$$

Proposition 2.1 (Compatibility of restrictions). *If $U \subseteq V \subseteq W$, then*

$$\text{res}_{U,U} = \text{id}_{\mathcal{O}(U)} \quad \text{and} \quad \text{res}_{W,U} = \text{res}_{V,U} \circ \text{res}_{W,V} \quad \text{i.e.}$$

$$\begin{array}{ccc} \mathcal{O}(W) & \xrightarrow{\text{res}_{W,V}} & \mathcal{O}(V) \\ & \searrow \text{res}_{W,U} & \downarrow \text{res}_{V,U} \\ & & \mathcal{O}(U) \end{array}$$

commutes.

Proof. Immediate from ordinary restriction of functions: $(f|_V)|_U = f|_U$. □

Proposition 2.2 (Local identity). *Let $U = \bigcup_{i \in I} U_i$ be an open cover. Suppose $f, g \in \mathcal{O}(U)$ and $f|_{U_i} = g|_{U_i}$ for every i . Then $f = g$.*

Proof. For each $x \in U$, choose i such that $x \in U_i$. Then $f(x) = f|_{U_i}(x) = g|_{U_i}(x) = g(x)$. □

Say: This is the “at most one gluing” part. A global function is determined by its restrictions to a cover.

Proposition 2.3 (Gluability). *Let $U = \bigcup_{i \in I} U_i$. Suppose we are given $f_i \in \mathcal{O}(U_i)$ such that for every i, j , $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$. Then there exists $f \in \mathcal{O}(U)$ such that $f|_{U_i} = f_i$ for every i .*

Proof. Define $f(x) := f_i(x)$ for any i such that $x \in U_i$. This is well-defined because the f_i agree on overlaps. For every $x \in U$, choose i with $x \in U_i$. On the neighbourhood U_i , $f = f_i$, so f is smooth near every point, hence smooth on U . □

Corollary 2.4. *Compatible local smooth functions glue to a **unique** global smooth function.*

Proof. Existence is gluability; uniqueness is local identity. **Key point:** This is the basic local-to-global principle that the abstract definition of a sheaf will capture. □

Germ and stalks in the smooth example

Say: Now fix a point p . A function contains much more information than we need if we only care about its behaviour arbitrarily close to p .

Definition 2.5 (Germ). Fix $p \in X$. Consider pairs (U, f) with $p \in U$, and $f \in \mathcal{O}(U)$. Define $(U, f) \sim (V, g)$ if there exists an open neighbourhood $p \in W \subseteq U \cap V$ such that $f|_W = g|_W$. The equivalence class of (U, f) is the **germ of f at p** , denoted f_p .

Definition 2.6 (Stalk). The set of all germs at p is called the **stalk** of $\mathcal{O}$ at p (denoted by $\mathcal{O}_p$). For $p \in U$ there is a canonical map $\mathcal{O}(U) \longrightarrow \mathcal{O}_p$, with $f \mapsto f_p$.

Remark 2.7. The statement $f_p = g_p$ means that f and g agree on some neighbourhood of p . It is much stronger than merely $f(p) = g(p)$. For example,

$$f(x) = x, \quad g(x) = x + x^2$$

have the same value at 0, but different germs at 0.

Proposition 2.8. $\mathcal{O}_p$ is naturally a commutative ring.

Proof. Define

$$[f]_p + [g]_p := [f + g]_p, \quad [f]_p [g]_p := [fg]_p,$$

after restricting representatives to a common neighbourhood of p .

If we change representatives, the new representatives agree with the old ones on sufficiently small neighbourhoods of p , so the resulting sums and products agree on a sufficiently small common neighbourhood. Hence the operations are well-defined. $\square$

Proposition 2.9 (The smooth stalk is local). Define $\mathfrak{m}_p := \{[f]_p \in \mathcal{O}_p : f(p) = 0\}$. Then $\mathcal{O}_p$ is a local ring with unique maximal ideal $\mathfrak{m}_p$, and $\mathcal{O}_p/\mathfrak{m}_p \cong \mathbb{R}$.

Proof. Evaluation at p defines a surjective ring homomorphism

$$\text{ev}_p : \mathcal{O}_p \longrightarrow \mathbb{R}, \quad [f]_p \mapsto f(p).$$

Its kernel is $\mathfrak{m}_p$, so $\mathcal{O}_p/\mathfrak{m}_p \cong \mathbb{R}$. Hence $\mathfrak{m}_p$ is maximal.

To prove uniqueness, let $[f]_p \notin \mathfrak{m}_p$. Then $f(p) \neq 0$. By continuity, f is nonzero on some open neighbourhood V of p . Hence $\frac{1}{f} \in C^\infty(V, \mathbb{R})$, and $[f]_p \left[\frac{1}{f} \right]_p = 1$. Thus every element outside $\mathfrak{m}_p$ is a unit. $\square$

Key point. What's the point? The stalk records functions “near p ”, and its maximal ideal records functions vanishing at p .

Observation. Since $\mathcal{O}_p/\mathfrak{m}_p \cong \mathbb{R}$, the quotient $\mathfrak{m}_p/\mathfrak{m}_p^2$ is naturally a real vector space. For a smooth manifold there is a natural isomorphism

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \xrightarrow{\sim} T_p^*X, \quad [f]_p + \mathfrak{m}_p^2 \mapsto d_p f.$$

Say: The point is that the familiar cotangent space can be recovered entirely from the local ring of functions. We will later see algebraic definition of the cotangent space of a scheme.

Presheaves

Say: Now forget that the sections are functions and keep only the restriction data.

Definition 2.10 (Presheaf). A **presheaf of sets** $\mathcal{F}$ on a topological space X consists of:

1. for every open set $U \subseteq X$, a set $\mathcal{F}(U)$;
2. for every inclusion $U \subseteq V$, a restriction map $\text{res}_{V,U} : \mathcal{F}(V) \longrightarrow \mathcal{F}(U)$, such that
 - $\text{res}_{U,U} = \text{id}_{\mathcal{F}(U)}$
 - for $U \subseteq V \subseteq W$, $\text{res}_{W,U} = \text{res}_{V,U} \circ \text{res}_{W,V}$.

Terminology. Elements of $\mathcal{F}(U)$ are called **sections of $\mathcal{F}$ over U** . Sections over all of X , $\mathcal{F}(X)$, are called **global sections**.

Proposition 2.11. *Let $\text{Open}(X)$ be the category whose objects are open subsets of X and whose morphisms are inclusions. Then a presheaf of sets on X is precisely a contravariant functor*

$$\mathcal{F} : \text{Open}(X)^{\text{op}} \longrightarrow \mathbf{Sets}.$$

Proof. An inclusion $U \subseteq V$ gives a morphism $U \rightarrow V$ in $\text{Open}(X)$, while the presheaf gives a map in the opposite direction $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$. The two restriction axioms are precisely preservation of identities and composition. $\square$

General germs and stalks

Fix $p \in X$. Let $\mathcal{N}_p$ be the category whose objects are open neighbourhoods $U \ni p$, with

$$U \longrightarrow V \iff V \subseteq U.$$

Thus restriction gives maps $\mathcal{F}(U) \longrightarrow \mathcal{F}(V)$.

Definition 2.12 (Stalk). The **stalk of $\mathcal{F}$ at p** is the filtered colimit

$$\mathcal{F}_p := \varinjlim_{U \in \mathcal{N}_p} \mathcal{F}(U).$$

Why filtered? If $U, V \ni p$, then $p \in U \cap V$ and there are arrows $U \longrightarrow U \cap V$, and $V \longrightarrow U \cap V$.

Remark 2.13 (Important warning). For smooth functions, the expression $f(p)$ makes sense. For an arbitrary sheaf or presheaf, a section need not have a “value” at p . What always makes sense is its germ $s_p \in \mathcal{F}_p$.

Sheaves

Definition 2.14 (Sheaf). A presheaf $\mathcal{F}$ is a **sheaf** if it satisfies the two axioms.

1. **Identity:** $U = \bigcup_{i \in I} U_i$. If $s, t \in \mathcal{F}(U)$, and $s|_{U_i} = t|_{U_i}$ for every i , then $s = t$. “at most one gluing”
2. **Glability:** Take $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for every i, j . Then $\exists s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i . “at least one gluing”

Key point. Compatible local sections glue to a unique global section.

Examples and a non-example

Example 2.15 (Functions). The following are sheaves:

$$U \mapsto C^\infty(U, \mathbb{R}), \quad U \mapsto C^0(U, \mathbb{R}),$$

and on a complex manifold, $U \mapsto \mathcal{O}_{\text{hol}}(U)$.

Example 2.16 (Non-example: A presheaf which is not a sheaf). Define

$$\mathcal{F}(U) = \{\text{bounded continuous functions } U \rightarrow \mathbb{R}\}.$$

Restriction preserves boundedness, so $\mathcal{F}$ is a presheaf. Now take $U = \mathbb{R}$, with $U_n = (-n, n)$, and define $f_n(x) = x$ on U_n . Each f_n is bounded on U_n , and the f_n agree on overlaps. If $\mathcal{F}$ were a sheaf, they would glue to a bounded continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$. But the only possible gluing is $f(x) = x$ which is unbounded. Hence gluing fails.

Example 2.17 (Skyscraper sheaf). Fix $p \in X$ and a set S . Define

$$\mathcal{S}(U) = \begin{cases} S, & p \in U, \\ \{*\}, & p \notin U. \end{cases}$$

The restriction maps are:

- id_S if both opens contain p ;
- the unique map to $\{*\}$ otherwise.

Then $\mathcal{S}$ is a sheaf, called the **skyscraper sheaf supported at p** .

Say: The point of this example is simply that a sheaf need not look like a familiar class of functions. All the information can be concentrated at one point.

The espace étalé viewpoint

Say: So far a sheaf is a large contravariant collection of sets. There is another geometric way to package exactly the same local information: put the stalk over each point.

Definition 2.18 (Espace étalé). Let $\mathcal{F}$ be a presheaf of sets on X . Define $E(\mathcal{F}) := \bigsqcup_{x \in X} \mathcal{F}_x$. An element may be written as (x, ξ) , for $\xi \in \mathcal{F}_x$. There is a natural projection

$$\pi : E(\mathcal{F}) \longrightarrow X, \quad (x, \xi) \longmapsto x.$$

Thus $\pi^{-1}(x) = \mathcal{F}_x$.

Sections give subsets of $E(\mathcal{F})$. If $s \in \mathcal{F}(U)$, then for every $x \in U$ we have a germ $s_x \in \mathcal{F}_x$. Hence s determines a map

$$\tilde{s} : U \longrightarrow E(\mathcal{F}), \quad x \longmapsto (x, s_x).$$

Write $\tilde{s}(U) = \{(x, s_x) : x \in U\}$.

Proposition 2.19. *The subsets $\tilde{s}(U)$, with $U \subseteq X$ open, $s \in \mathcal{F}(U)$ form a basis for a topology on $E(\mathcal{F})$.*

Proof. First, every $(x, \xi) \in E(\mathcal{F})$ lies in one of these sets because $\xi \in \mathcal{F}_x$ is represented by some section $s \in \mathcal{F}(U)$ on a neighbourhood $U \ni x$.

Now suppose $(x, \xi) \in \tilde{s}(U) \cap \tilde{t}(V)$. Then $s_x = \xi = t_x$. Equality of germs means there exists an open neighbourhood $x \in W \subseteq U \cap V$ such that $s|_W = t|_W$. Hence $\tilde{s}(W) = \tilde{t}(W) \subseteq \tilde{s}(U) \cap \tilde{t}(V)$, and this basic open contains (x, ξ) . $\square$

Proposition 2.20. *For every $s \in \mathcal{F}(U)$, $\pi|_{\tilde{s}(U)} : \tilde{s}(U) \rightarrow U$ is a homeomorphism.*

Proof. It is bijective, with inverse $x \mapsto (x, s_x)$. The map $\pi|_{\tilde{s}(U)}$ is continuous because for an open $V \subseteq X$,

$$\pi^{-1}(V) \cap \tilde{s}(U) = \widetilde{s|_{U \cap V}}(U \cap V),$$

which is open in $\tilde{s}(U)$. For continuity of the inverse, take a basic open $\tilde{t}(V)$. Its preimage in U is $\{x \in U \cap V : s_x = t_x\}$. If $s_x = t_x$, then the two sections agree on some neighbourhood of x , so this set is open. $\square$

Corollary 2.21. *The projection $\pi : E(\mathcal{F}) \rightarrow X$ is a local homeomorphism. i.e. The stalk $\mathcal{F}_x$ becomes the literal fibre over x .*

Proposition 2.22 (For a sheaf). *If $\mathcal{F}$ is a sheaf, then for every open $U \subseteq X$,*

$$\mathcal{F}(U) \cong \{\sigma : U \rightarrow E(\mathcal{F}) : \sigma \text{ continuous and } \pi \circ \sigma = \text{id}_U\}.$$

Thus sections of the sheaf are exactly continuous sections of the espace étalé.

Proof sketch. Every $s \in \mathcal{F}(U)$ gives the continuous section $\tilde{s}(x) = (x, s_x)$.

Conversely, let $\sigma : U \rightarrow E(\mathcal{F})$ be a continuous section of π . Since π is a local homeomorphism, every point $x \in U$ has a neighbourhood U_x on which σ is of the form $y \mapsto (y, (s_x)_y)$ for some $s_x \in \mathcal{F}(U_x)$. On overlaps these local sections have the same germs at every point, hence by the sheaf identity axiom they agree locally. The sheaf gluing axiom then gives a unique global $s \in \mathcal{F}(U)$ with $\tilde{s} = \sigma$. $\square$

Remark 2.23. For a general presheaf $\mathcal{F}$, the sheaf of continuous sections of

$$E(\mathcal{F}) \rightarrow X$$

need not recover $\mathcal{F}$ itself. It recovers the **sheafification** of $\mathcal{F}$, which we will see in the future.

Key point. A sheaf of sets can be viewed equivalently as an étalé space, i.e. a local homeomorphism over X .

Ringed spaces

Say: Return to the original example. $C^\infty(U)$ was not merely a set; it was a ring. In geometry we usually want to remember this algebraic structure.

Definition 2.24 (Ringed space). A **ringed space** is a pair $(X, \mathcal{O}_X)$, where X is a topological space and $\mathcal{O}_X$ is a sheaf of rings on X .

- The sheaf $\mathcal{O}_X$ is called the **structure sheaf**.
- For $U \subseteq X$ open, elements of $\mathcal{O}_X(U)$ are called **functions on U** , even when they are not literally ordinary set-theoretic functions.

Example 2.25. If X is a smooth manifold, then $\mathcal{O}_X(U) = C^\infty(U, \mathbb{R})$ defines a ringed space $(X, \mathcal{O}_X)$.

Remark 2.26. If $U \subseteq X$ is open, then the restriction $\mathcal{O}_X|_U$ is a sheaf of rings on U . We usually denote it by $\mathcal{O}_U$. Thus $(U, \mathcal{O}_U)$ is itself a ringed space.

$\mathcal{O}_X$ -modules

Say: Once each open set carries a ring $\mathcal{O}_X(U)$, the next obvious question is: what is the sheaf-theoretic analogue of a module over a ring?

Definition 2.27 ($\mathcal{O}_X$ -module). Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$ -module $\mathcal{F}$ is a sheaf of abelian groups such that:

1. for every open $U \subseteq X$, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module;
2. these module structures are compatible with restriction: if $U \subseteq V$, $f \in \mathcal{O}_X(V)$, and $s \in \mathcal{F}(V)$, then $(fs)|_U = (f|_U)(s|_U)$.

Equivalently, for every $U \subseteq V$ the diagram

$$\begin{array}{ccc} \mathcal{O}_X(V) \times \mathcal{F}(V) & \longrightarrow & \mathcal{F}(V) \\ \text{res} \times \text{res} \downarrow & & \downarrow \text{res} \\ \mathcal{O}_X(U) \times \mathcal{F}(U) & \longrightarrow & \mathcal{F}(U) \end{array}$$

commutes.

Example 2.28. The structure sheaf $\mathcal{O}_X$ is naturally an $\mathcal{O}_X$ -module over itself.

Example 2.29 (Vector bundles). Let X be a smooth manifold and $\pi : E \rightarrow X$ a smooth vector bundle. Let $\mathcal{E}(U) = \{\text{smooth sections } U \rightarrow E\}$. Then $\mathcal{E}$ is a sheaf, and it is naturally an $\mathcal{O}_X$ -module by pointwise multiplication:

$$(fs)(x) = f(x)s(x), \quad f \in C^\infty(U), \quad s \in \mathcal{E}(U).$$

Say: This is the model to keep in mind: vector bundles give locally free $\mathcal{O}_X$ -modules. Later, in algebraic geometry, the sheaf-theoretic object will be primary.

Proposition 2.30. If $\mathcal{F}$ is an $\mathcal{O}_X$ -module, then for every $p \in X$, the stalk $\mathcal{F}_p$ is naturally an $\mathcal{O}_{X,p}$ -module.

Proof. Take $[f]_p \in \mathcal{O}_{X,p}$, and $[s]_p \in \mathcal{F}_p$. Choose representatives $f \in \mathcal{O}_X(U)$, and $s \in \mathcal{F}(V)$ with $p \in U \cap V$. Restrict both to $W = U \cap V$ and define

$$[f]_p \cdot [s]_p := [(f|_W)(s|_W)]_p.$$

If we choose different representatives, then after restricting to a sufficiently small common neighbourhood they agree, so the product germ is unchanged. Hence the operation is well-defined, and the module axioms are inherited from the module structures on sections. $\square$

Key point.

$$\text{ring } A \leftrightarrow \text{structure sheaf } \mathcal{O}_X, \quad A\text{-module } M \leftrightarrow \mathcal{O}_X\text{-module } \mathcal{F},$$

and stalks give the local algebra

$$\mathcal{O}_{X,p}, \quad \mathcal{F}_p.$$